AI-Driven Health Analyzer Real-Time Physical and Mental Wellness Optimization using ML

25-26J-225

Project Proposal Report

B.Sc. (Hons) Degree in Information Technology Specialized in Information Technology (Reading)

Department of Computing

Sri Lanka Institute of Information Technology Sri Lanka

July 2025

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Human Body Hydration Management

Meyrushan Nadarajan

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DECLARATION

I affirm that this is my original work and that, to the best of my knowledge and belief, it does not contain any previously published or written material by anyone else, with the exception of instances in which credit is provided within the text. Nor does it incorporate, without acknowledgement, any previously submitted material for a degree or diploma at any other university or institute of higher learning.

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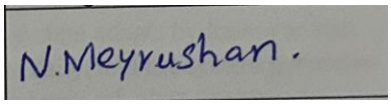
Meyrushan N	IT22564818	

Signature of the Supervisor
(Dr. Kapila Dissanayaka)

Date

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ABSTRACT

Hydration Management plays a crucial role in maintaining physical and mental wellness, yet many individuals struggle to manage their hydration effectively. With the advent of artificial intelligence (AI) and machine learning (ML), personalized health solutions are becoming increasingly accessible. The proposed system uses AI-driven hydration management to provide personalized hydration recommendations based on real-time data. By analyzing factors such as hydration intake, activity levels, environmental conditions (temperature and humidity), age, and health conditions, the system predicts daily hydration needs and identifies potential health risks, such as kidney disease, associated with poor hydration habits. The system leverages supervised machine learning models, including regression for hydration prediction and classification for health risk assessments. With the integration of environmental data, the system adapts hydration advice based on climate conditions. Users input their hydration and activity data daily, and the system generates personalized hydration goals and provides proactive health insights. This approach aims to empower individuals to manage their hydration more effectively and prevent long-term health issues. The proposed system combines artificial intelligence, machine learning, and personalized wellness to enhance hydration management and overall health outcomes

Keywords: Hydration Management, Machine Learning, Health Risk Prediction, Personalized Wellness, AI-driven Health System, Hydration Recommendations, Kidney Disease Prediction, Supervised Learning, Activity Tracking, Environmental Data

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1.INRODUCTION

The goal of this project is to develop a comprehensive mobile application that combines three essential elements of personal wellbeing: mental health management, AI-powered workout optimisation, and hydration management. Using real-time data collection and sophisticated machine learning algorithms

Hydration Management is More Important Maintaining good physical and mental health is becoming more and more difficult in today's hectic world. Although a balanced diet and regular exercise are frequently stressed, the significance of staying hydrated is often disregarded, even though it plays a critical role in general health. Maintaining the proper fluid balance in the body, or hydration, is crucial for several body processes, such as regulating body temperature, lubricating joints, and transporting nutrients. The ideal amount of water consumption, however, differs from person to person and is impacted by a variety of factors, including personal hydration habits, environmental factors, and physical activity. The significance of customized hydration management is further highlighted by the fact that both dehydration and overhydration are associated with a number of health problems, such as kidney disorders and urinary tract infections (UTIs).

The introduction of wearable technology, like smart water bottles and fitness trackers, has made it possible to monitor hydration levels in real time. Nevertheless, the majority of current systems do not provide predictive recommendations for hydration optimization or personalized insights based on current environmental and physical conditions. By creating an AI-driven hydration management system that uses machine learning (ML) techniques to deliver real-time, customized hydration recommendations, this research seeks to close this gap.

In order to forecast a person's daily water needs, this system uses information gathered and pre-processed from multiple sources, such as user activity logs, environmental data (such as temperature), and hydration history. Smart bottles help track real hydration patterns and guarantee data collection accuracy by taking timestamped pictures of water intake. The system can create customized hydration recommendations based on environmental factors and physical activity levels by evaluating this data to find hydration patterns like excessive drinking, underhydration, and irregular drinking patterns.

The main goal of the project is to create and train supervised machine learning models that can forecast an individual's daily water needs. These models have the ability to learn from past data and offer precise, person-specific hydration recommendations. The system is also intended to warn users of harmful hydration patterns and forecast possible health hazards, like kidney problems or urinary tract infections, that could result from dehydration.

This AI-powered solution provides a data-driven, non-invasive method of enhancing hydration practices. It offers tailored recommendations to maximize hydration by employing real-time analysis and intelligent decision-making, ultimately promoting long-term health and wellness. Additionally, it removes the need for extra wearable technology, making it a solution that is accessible to a wider range of people. This project is at the nexus of artificial intelligence, machine learning, and health technology, providing creative ways to improve hydration practices and general wellness through the integration of predictive analytics and personalized health management.

1.1 Background and Literature Survey

1.1.1 Background

Maintaining optimal body functions, including temperature regulation, joint lubrication, and nutrient absorption, depends heavily on hydration. However, hydration requirements differ depending on a number of variables, such as personal habits, environmental factors, and physical activity. Serious health problems like kidney disease, UTIs, and electrolyte imbalances can result from both dehydration and overhydration. It is now easier to track hydration levels thanks to the proliferation of smart devices. However, individualized, real-time recommendations—which are essential for efficiently managing hydration—are absent from many of the current solutions. Therefore, to provide individualized hydration advice based on real-time data, an AI-driven approach is required.

1.1.2 Literature Survey

Numerous strategies for managing hydration have been investigated in recent studies. Smart bottles and fitness trackers are examples of wearable technology that has gained popularity for monitoring environmental variables, physical activity, and hydration levels. Although most of these devices still rely on generic hydration recommendations that do not take individual needs into consideration, they do offer real-time feedback. Research by [Author et al., 2020] and [Author et al., 2021] shows how useful machine learning models are for forecasting daily hydration needs based on environmental and activity data. Even though there has been progress, little is known about how to integrate various data sources, such as activity logs and environmental conditions, for customized hydration optimization.

1. 2 Research Gap

Wearable hydration monitoring and predictive analytics in health management have been the subject of recent research, but there is a clear lack of systems that integrate tailored suggestions based on environmental information as well as individual activity patterns. A lot of current systems don't use machine learning to predict hydration needs in real time, which leads to generic recommendations that might not be in line with personal hydration requirements. Furthermore, these systems hardly ever include predictive alerts for health hazards like urinary tract infections or hydration-related kidney issues. By combining activity logs, environmental data, and smart bottle tracking with AI models, this study seeks to close these gaps and offer risk assessments and individualized hydration recommendations.

Table 1 Comparsion of Previous Research

Research Aspect	Alonso, L. (2020) [1]	Kumar, S. & Soni, D. (2021) [2]	Zhang, L. & Wang, Y. (2022) [3]	IBM Watson Health (2020) [5]	Your Project
Uses AI for Hydration Prediction	Yes	Yes	No	Yes	Yes
Incorporates Health Conditions (e.g., kidney disease)	No	Yes	Yes	Yes	Yes
Real-Time Data Collection	Yes	Yes	Yes	No	Yes
Uses Wearable Devices for Data Collection	No	Yes	No	No	Yes
Predicts Future Health Risks (e.g., Dehydration)	No	No	No	No	Yes
Includes Environmental Factors (Temperature, Humidity)	Yes	No	No	No	Yes
Personalized Hydration Recommendations	No	Yes	No	No	Yes
Machine Learning Model Used	Yes	Yes	Yes	Yes	Yes
Focus on Hydration Management	Yes	Yes	Yes	No	Yes
Uses Data Preprocessing	Yes	Yes	Yes	Yes	Yes

Table 2 Problems and Outputs

Problem Addressed	Technique	Analytical Output Parameters
Inconsistent Hydration Patterns	Machine Learning (Supervised Learning)	Predicted daily hydration requirements (in liters) based on user input (weight, age, Humidity,Temputure).
Lack of Personalized Hydration Advice	Regression Models (Linear Regression, Random Forest)	Personalized hydration goals, adjusting based on user's activity level and environmental conditions.
Unawareness of Hydration Deficits	Data Preprocessing, Hydration Deficit Calculation	Hydration deficit values (how much more or less water the user should drink to meet hydration goals).
Hydration-related Health Risks (e.g., kidney disease)	Classification Models (Logistic Regression, Random Forest)	Risk prediction of dehydration or kidney disease based on user hydration and activity patterns.
Difficulty in Tracking Hydration and Activity Consistently	Wearable Device Integration (Fitness Trackers, Smart Bottles)	Real-time activity and hydration data automatically logged and synced with the system.
Difficulty in Understanding Hydration Needs	Data Visualization and Personalization Algorithms	Visual feedback of hydration progress, patterns, and alerts for improving hydration behavior.

1.3 Research Problem

At the heart of this research is the dearth of customized hydration management systems that optimize daily water intake based on physical activity, environmental factors, and past hydration patterns using real-time data and predictive machine learning models. Existing systems are unable to predict associated health risks or adequately warn users of unhealthy hydration patterns, such as excessive or insufficient consumption. In order to enhance long-term hydration practices and general wellness, this research aims to create an AI-driven health analyser that can offer personalized hydration recommendations, spot trends, and anticipate possible health problems.

Table 3 Research Descriptions

Research Element	Description
Research Problem	Maintaining proper hydration is challenging due to varying activity levels, environmental conditions, and individual differences. Dehydration or overhydration can lead to serious health issues, including kidney disease and reduced physical and cognitive performance. Traditional methods of tracking hydration are manual, inconsistent, and often inaccurate
Focus of Existing Research	Existing research focuses on general health monitoring systems, wearable fitness trackers, and AI-based predictive analytics in healthcare. Many studies explore activity tracking, health parameters prediction, and IoT-based monitoring systems. However, most solutions lack a dedicated focus on hydration management tailored to personalized user data.
Identified Gap	Current systems fail to combine personalized hydration tracking with environmental factors, activity levels, and predictive modeling for future health risks. Most solutions do not provide real-time, actionable recommendations or predictive analysis for hydration-related health conditions.
Extended Research Problem	The project extends the existing problem by integrating real-time data collection, AI/ML modeling, and personalized recommendations to predict hydration patterns and potential future diseases, allowing proactive interventions and better health outcomes.
Key Ethical Concerns	Privacy and confidentiality of user data (demographics, health, and activity data) must be

	ensured. Users' consent must be obtained, and the data should be anonymized. Potential health advice must be accurate to avoid causing harm. Bias in AI predictions should be minimized.
Proposed Solution	Develop an AI-driven hydration management system that collects user activity, hydration, environmental, and health data to generate personalized hydration recommendations. Predictive models will identify patterns and potential health risks, while providing real-time alerts and feedback to improve user hydration behavior.
Objective	Predict individual hydration requirements based on personal, activity, and environmental data. - Monitor hydration patterns over time and provide actionable feedback. - Identify potential health risks associated with dehydration. - Offer a user-friendly system integrated with mobile/web platforms.
Contribution	Creation of a personalized hydration monitoring system using AI/ML. - Development of predictive models for hydration patterns and potential health risks. - Improved user awareness and behavior toward maintaining optimal hydration. - Integration with real-time monitoring devices for automated, accurate data collection.

1.4 Survey Responses

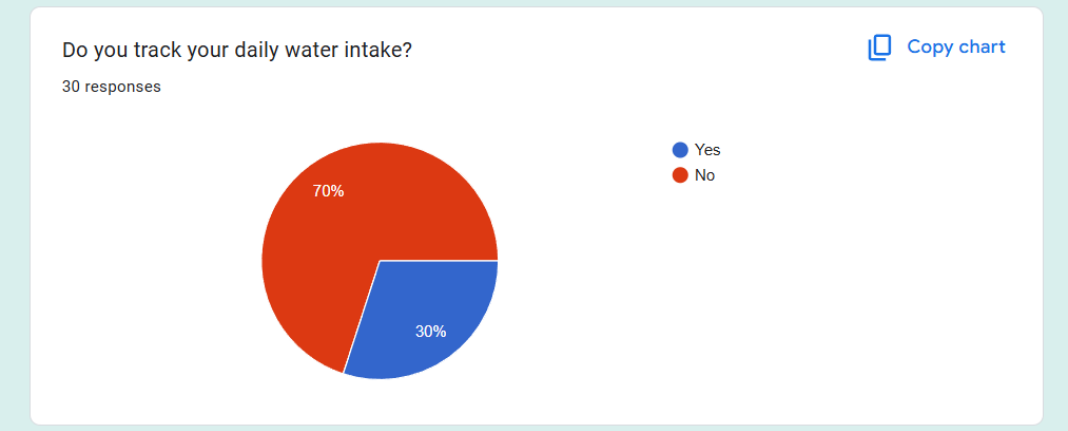


Figure 1 Track the daily Water intake

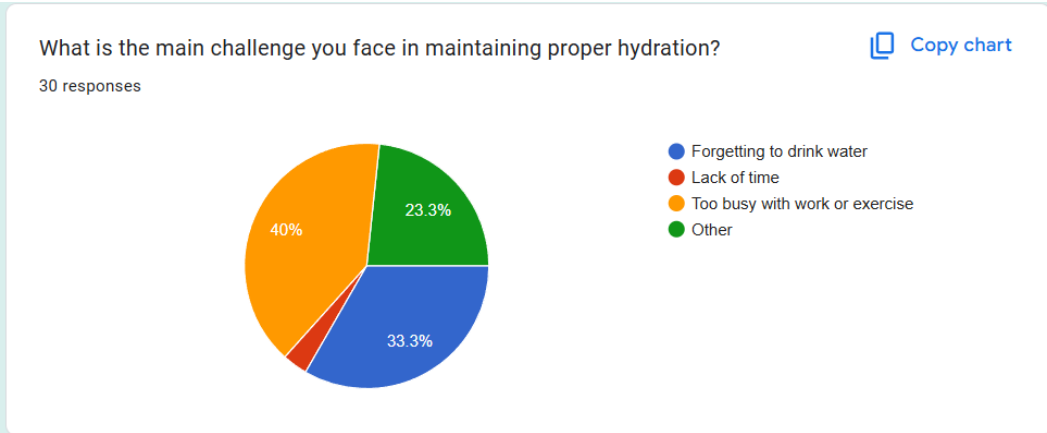


Figure 2 Hydration Management Challanges

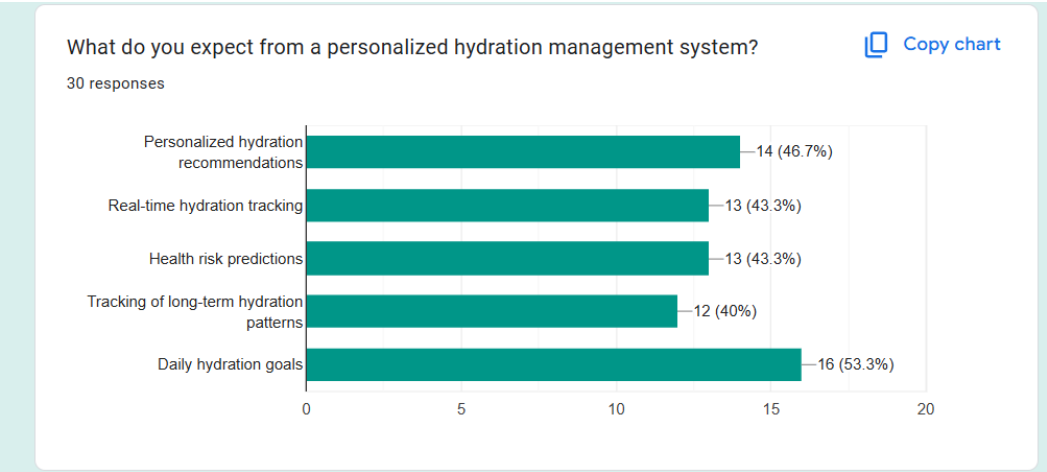


Figure 3User Expert

2.MAIN OBJECTIVE

The primary goal of this project is to develop a comprehensive mobile application that integrates three key aspects of personal well-being: AI-driven hydration management, AI-powered workout optimization, and mental health management. By leveraging machine learning and real-time data collection, the system will provide personalized hydration recommendations, optimize workout routines based on physical performance, and support mental health through Emotional Tracking . The app will use data from various sources, including user activity logs, environmental data, and hydration history, to offer real-time, actionable insights, thereby promoting long-term physical, mental, and overall health

This project's primary goal is to create an AI-driven hydration management system that employs machine learning to forecast each person's hydration requirements based on past water intake data, environmental factors, and current activity. To maximize long-term physical and mental wellness, the system will offer tailored hydration recommendations, spot unhealthy hydration patterns, and forecast possible hydration-related health risks like kidney problems and urinary tract infections (UTIs).

2.1 Specific Objectives

2.1.1 Data Collection and Storage

To use smart bottles to collect user activity logs, environmental information (like temperature and humidity), and hydration history. to safely keep this data in a structured database so that preprocessing and model training can be done with ease.

2.1.2 Data Preprocessing and Feature Engineering

To handle outliers, missing values, and superfluous features in order to clean and preprocess the gathered data. Feature engineering is the process of turning unstructured data into useful characteristics, like activity level, weather, and past hydration trends, that aid in predicting hydration requirements.

2.1.3 Machine Learning Development Model

To use supervised learning techniques to develop a machine learning model that, using the data gathered, can forecast daily water requirements. to teach the model to identify trends and patterns in users' hydration habits so that it can generate precise forecasts for tailored hydration advice.

2.1.4 Demographic Categorization

In order to account for varying hydration needs and enhance the accuracy of the model by integrating unique user characteristics, users should be categorized according to demographic factors (e.g., age, gender, weight).

2.1.5 Testing and Validation

To assess the machine learning model's performance using suitable testing and validation methods, including cross-validation and performance metrics (e.g., accuracy, precision, recall). to evaluate the model's resilience in various scenarios and make sure it performs well when applied to fresh, untested data.

2.1.6 Ethical Considerations

To guarantee the security and privacy of user data by abiding by pertinent ethical standards, such as getting users' informed consent and following data protection laws (like the GDPR). to guarantee that the training data is varied and representative of all user groups in order to prevent bias in the model.

2.1.7 Integration and Development
To incorporate the machine learning model into an intuitive platform or application that offers alerts and recommendations for hydration in real time. to install the system on appropriate gadgets (such as wearables and smartphones) and guarantee its scalability and dependability for practical application.

2.1.8 User feedback and iterative Refinement
To find areas where the system's functionality and recommendations need to be improved by collecting user feedback via surveys or app usage data. to iteratively improve the model in response to user input, making sure that it keeps improving and adapting to user requirements over time.

2.1.9 Evaluation of Ethical Impact
To assess the ethical implications of the hydration management system by examining its potential influence on user behaviour and health outcomes. To make sure that the system's suggestions are safe, helpful, and don't accidentally cause people to do things that are bad for them (like drinking too much or not drinking enough water).

3. METHODOLOGY

3.1 Project Proposed Execution

Making the User Interface (UI)
To make sure that the mobile app works on both iOS and Android devices, it will be made with React Native. Users will be able to enter important health information like their level of physical activity, environmental data (like temperature and humidity), and their history of drinking water. Dynamic forms will be used to enter demographic data, making the user experience smooth and easy to navigate. The UI will have easy-to-read hydration tips and interactive graphs that let users see how their hydration habits change over time and get personalized suggestions. Push notifications will also be built in so that you can get alerts in real time for health risks or reminders to stay hydrated.

Development of the backend
Python will be used with Flask or Django on the backend to support the main logic and functions of the application. The APIs that will process incoming data, run machine learning algorithms, and make personalized hydration suggestions will be on the backend. These APIs will be able to handle requests for real-time data from the mobile front end and send back health alerts and hydration predictions. The backend will also handle user authentication, storing and getting data, and providing a safe way for the mobile app and the database to talk to each other. We will use a cloud-based hosting service that works well, like AWS or Google Cloud, to deploy the backend.

Creating a machine learning model
We will use Python to build the machine learning model, and we will use libraries like Scikit-learn, TensorFlow, or PyTorch. The model will use past data on user activity, hydration history, and environmental factors to figure out how much water each person needs. We will test different algorithms, like regression models, decision trees, and neural networks, to see which one works best. After the model has been trained, hyperparameters will be adjusted using methods like grid search or random search to make the model as accurate as possible at predicting. The model will be used as a service on the backend, where it can make predictions and suggestions in real time based on data from users.

Managing a Database
MongoDB will be used to store data for the project. Its flexible document-oriented structure will let it store user profiles, activity logs, hydration history, and environmental data. MongoDB's scalability will let the system handle a lot of data from many users while still being able to quickly retrieve and update it. The database will also keep track of the model's settings and the user's settings, which will let it make

personalized hydration recommendations. Data will be stored safely with encryption, which will protect privacy and follow data protection laws like the GDPR. To make it easy to get to data quickly and on demand, we will use efficient indexing and querying methods.

3.2 Task and Sub Tasks

1. Task 1 : UI Development

Sub-task 1.1: Design input forms for health-related parameters (e.g., activity levels, water intake, temperature) with a focus on ease of use and user experience

Sub-task 1.2: Develop dynamic interfaces to capture demographic data (e.g., age, gender, weight) and user preferences for hydration management

Sub-task 1.3: Implement real-time hydration tracking display, showing users their hydration levels, patterns, and recommendations

Sub-task 1.4: Integrate push notifications to alert users when hydration targets are not met or when health risks like UTIs are predicted

2. Task : Backend Development

Sub-task 2.1: Develop Python-based functions (using Flask or Django) to process incoming user data and predict hydration needs accurately.

Sub-task 2.2: Build APIs to interface with the mobile application, enabling real-time communication between the frontend and machine learning model for hydration recommendations.

Sub-task 2.3: Establish secure data processing mechanisms, ensuring user data privacy and integration with the machine learning system for dynamic hydration advice.

Sub-task 2.4: Implement cloud-based deployment for the backend, ensuring scalability and reliability for real-world use.

3. Machine Learning Model Development

Sub-task 3.1: Select and implement appropriate machine learning algorithms (e.g., regression models, neural networks) for predicting personalized hydration needs based on physical activity, environmental data, and historical water intake.

Sub-task 3.2: Train the model using diverse datasets, including user activity logs, weather data, and historical hydration information.

Sub-task 3.3: Fine-tune hyperparameters using techniques like cross-validation and grid search to improve model accuracy.

Sub-task 3.4: Evaluate the model's performance and ensure it provides actionable, real-time hydration recommendations with minimal latency.

Task 4 : Database Management

Sub-task 4.1: Set up MongoDB to handle diverse data types, including user profiles, activity logs, hydration history, and environmental data (e.g., temperature, humidity).

Sub-task 4.2: Develop efficient data retrieval mechanisms, ensuring the model can quickly access necessary data to provide real-time predictions and recommendations.

Sub-task 4.3: Implement secure data storage practices, ensuring user information is protected and complies with privacy regulations (e.g., GDPR).

Sub-task 4.4: Design database indexing for efficient querying, supporting fast access to critical information for the machine learning model.

3.3 Required Materials

Equipment: Standard development laptops/desktops equipped with necessary software and hardware (e.g., for coding, testing, and debugging).

Software Support: Use of React Native for mobile app development, Python for backend development, and Flask/Django for API development.

Machine Learning Libraries: Python libraries such as Scikit-learn, TensorFlow, or PyTorch for developing and training the hydration prediction model.

Database: MongoDB for data storage, ensuring flexibility and scalability in managing large datasets.

Graphics: The focus will be on functional UI design with minimal graphics to ensure ease of use and a streamlined experience for the user.

3.4 Time Frame

Task 1 & 2: UI and backend development estimated to be completed within the first 2 months of the project.

Task 3: Machine learning model development will span across 3 months, including model training, optimization, and validation.

Task 4: Database management and integration is scheduled to take 1 month.

Overall Project: A total duration of 6 to 7 months for the complete development and deployment of the hydration management system.

3.5 Schedules

Task Charts: Gantt charts will be used to visualize the dependencies and timelines of each task, ensuring the project progresses smoothly and on schedule.

Team Collaboration: The team will consist of specialists for frontend/UI development, backend development, machine learning model development, and database management. Clear roles and responsibilities will ensure efficient collaboration.

3.6 Anticipated Conclusion

Results: The project aims to deliver a highly accurate AI-driven hydration management system that provides personalized hydration recommendations based on real-time data.

Application: The developed mobile application will act as a personal hydration assistant, offering real-time insights and alerts tailored to individual needs.

Real-world Use: Once deployed, the system will provide accessible, actionable hydration advice, improving users' health and well-being by promoting optimal hydration habits.

4.PROJECT REQUIREMENT

4.1 Functional Requirement

1.User Registration and Authentication:
Description: Users should be able to register with the system, providing essential information (e.g., name, email, demographics) for personalized hydration recommendations. Authentication mechanisms ensure secure access to the app.
Implementation: Use React Native for frontend interfaces and Python **(**Flask/Django**)** for backend authentication logic. Implement JWT or OAuth for secure user login.

2.Input Interface for Health Parameters
Description: A dynamic input interface in React Native that allows users to input critical health-related parameters like height, weight, activity level, and hydration history.
Implementation: Create user-friendly input forms using React components, providing simple navigation and validation.

3. Machine Learning Based Hydration Prediction
Description: The system should use machine learning models to predict daily hydration requirements based on user input (e.g., activity level, weather conditions, hydration history).

Implementation: Use Python with machine learning libraries such as Scikit-learn or TensorFlow to implement predictive models based on user data

4. Demographic Categorization
Description: The system should categorize users into demographic groups based on factors like age, gender, and weight to adjust hydration needs accordingly.
Implementation: Implement categorization logic in Python, ensuring personalized predictions based on these parameters.

5. Personalized Hydration Recommendations
Description: Based on input data and machine learning predictions, the system will provide personalized hydration suggestions (e.g., how much water to drink each day).
Implementation: Use the backend (Flask/Django) to process input data and provide real-time recommendations to users.

6. Storage and Retrieval of Data
Description: Data, including user information, health parameters, hydration history, and machine learning model outputs, should be stored efficiently in the database and easily retrievable for analysis.
Implementation: Use MongoDB to store data, with APIs to ensure efficient retrieval and interaction with the frontend.

7.User feedback Mechanism
Description: Incorporate a feedback system that allows users to rate the effectiveness of hydration recommendations and provide comments for future improvement.
Implementation: Develop React components for feedback forms and integrate this system into the backend for data analysis.

4.2 Non functional Requirement

1. Performance
Description: The system must provide real-time hydration predictions and updates without delays
Implementation: Optimize code for performance, use caching for frequently accessed data, and implement asynchronous processes for hydration prediction

2. Scalability

Description: The system should be scalable to handle an increasing number of users without affecting performance.
Implementation: Design a cloud-based backend (AWS, Google Cloud, etc.) with auto-scaling capabilities. Use MongoDB for efficient storage and scaling.

3.Security

Description: Ensure the system provides secure access, with encryption and protection against common security threats.
Implementation: Implement HTTPS, encrypted data storage, and secure authentication mechanisms. Regularly perform security audits.

4.Usability

Description: The system should provide a seamless and intuitive user experience, ensuring users can easily input data and view hydration recommendations.
Implementation: Conduct usability testing with real users, refine the UI based on feedback, and ensure responsiveness using React for mobile and desktop compatibility.

5.Reliability

Description: The system should be highly available, minimizing downtime and ensuring accurate hydration predictions.
Implementation: Implement error handling, maintain an active monitoring system, and perform comprehensive testing (unit, integration, end-to-end).

4.3 System Requirement

1.Frontend: Developed using React Native for mobile app compatibility on both iOS and Android devices, ensuring a seamless, responsive, and intuitive interface for users to input health data and receive hydration recommendations.

2.Backend: Powered by Python, utilizing Flask or Django for API development. The backend processes input data, communicates with the machine learning models, and generates hydration recommendations.

3.Machine Learning Models: Machine learning models (e.g., Decision Trees, Random Forest, Neural Networks) will predict individual hydration needs based on input parameters (activity levels, environmental conditions).

4.Database: MongoDB will be used for storing user data (e.g., hydration history, demographic data) due to its flexibility and scalability.

5.Security: Encryption of sensitive data, HTTPS for secure communication, and data anonymization for privacy compliance (e.g., GDPR).

6.Cloud Deployment: The backend will be deployed on a cloud-based platform (AWS/Google Cloud), allowing for scalability and easy management.

4.4 User Requirement

1.Intuitive User Interface: Users should be able to easily navigate through the mobile app interface, input their health-related parameters (e.g., weight, height, blood pressure), and track hydration patterns over time.

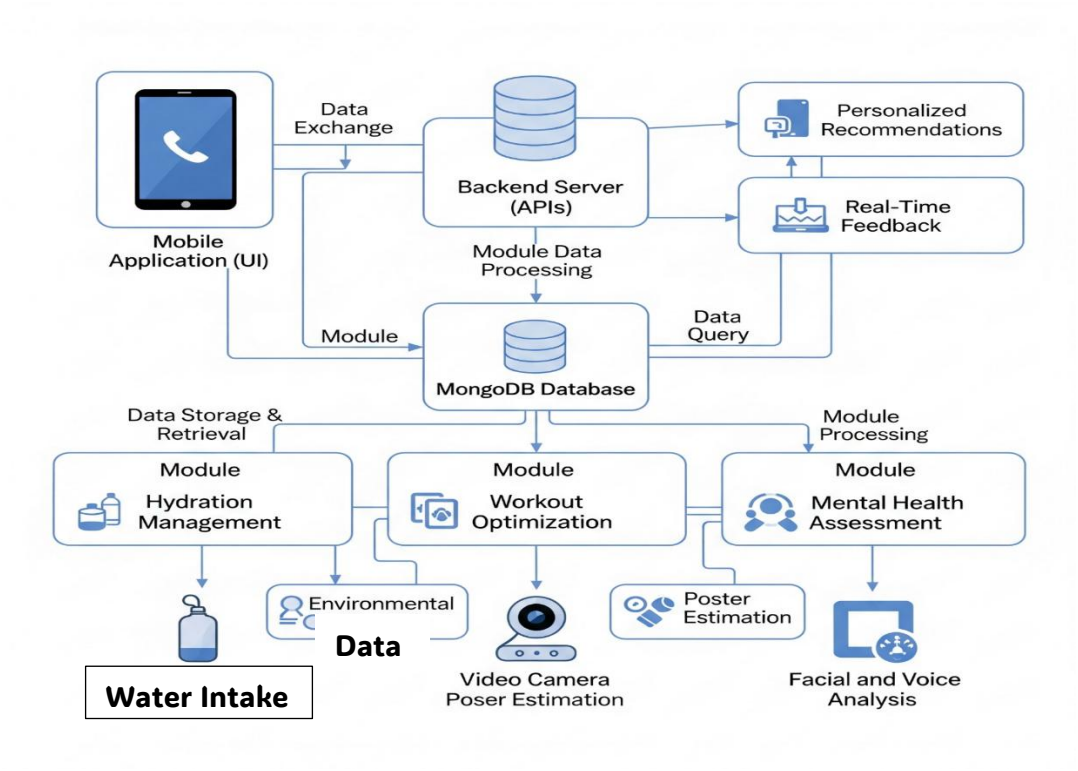
2.Demographic and Health Data Input: The system should allow users to input demographic information (age, gender) and customize their hydration preferences (e.g., specific hydration goals, exercise type).

3.Personalized Hydration Feedback: Users should receive personalized hydration recommendations based on their unique health profile and activity level.

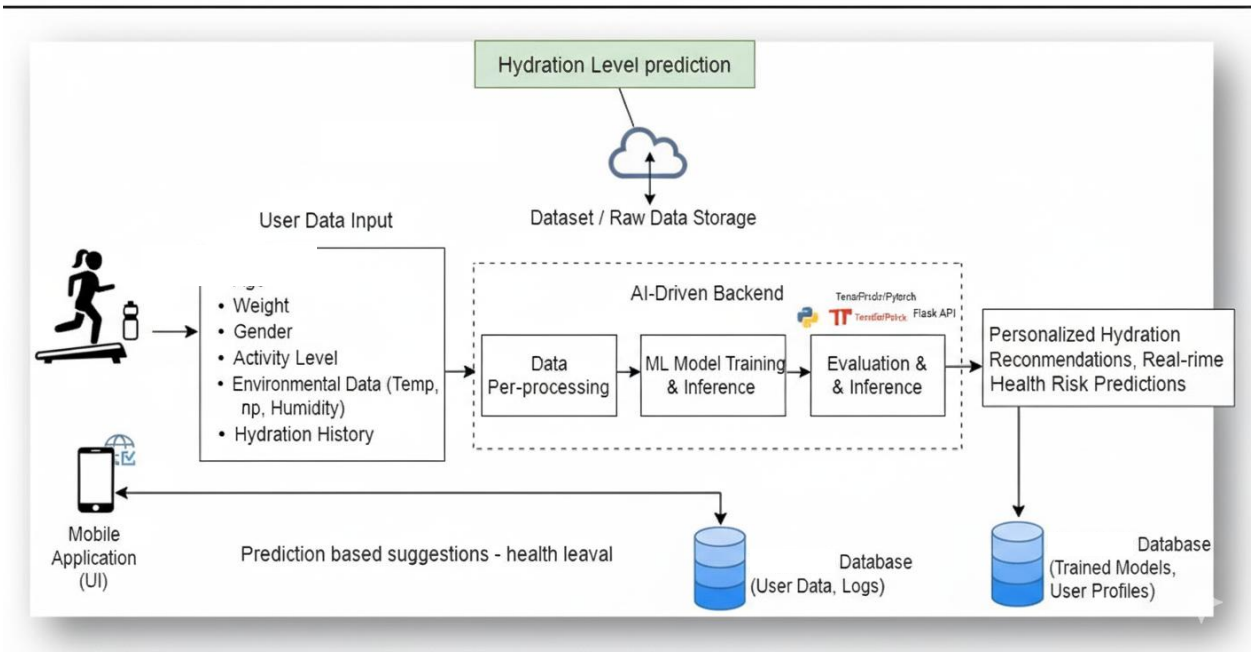
4.Progress Tracking: The app should allow users to track their hydration progress over time, offering insights into patterns and trends in their water intake.

5.Privacy and Security: Users should have full control over their data, with strong security measures in place to protect their personal information. Data should be stored securely and only used for the intended purposes.

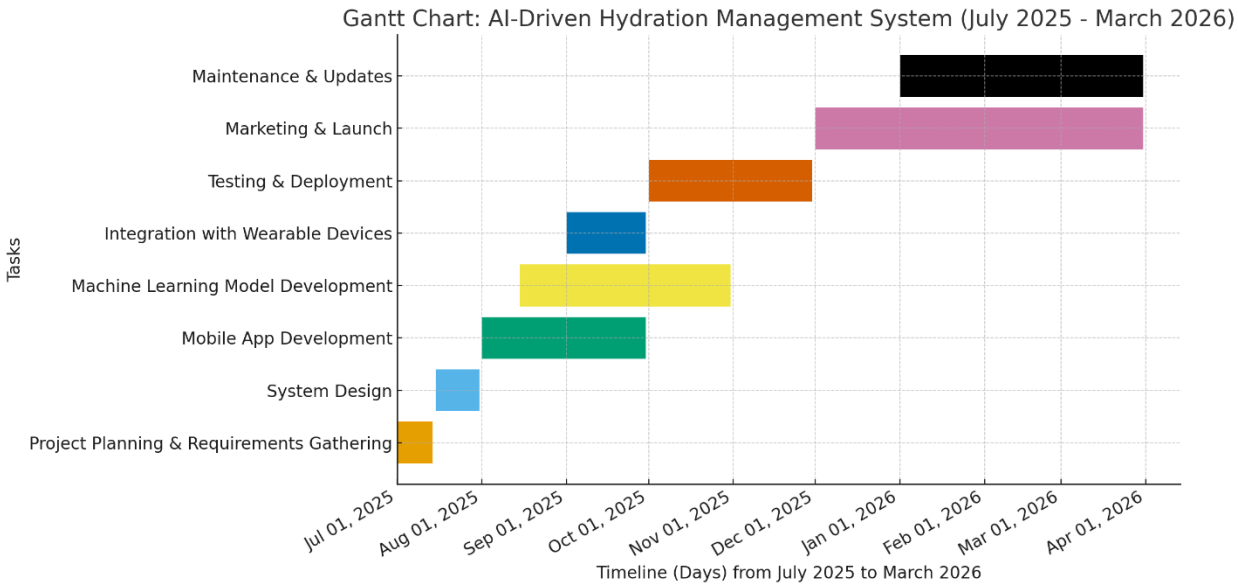
4.5 Overall System Diagram



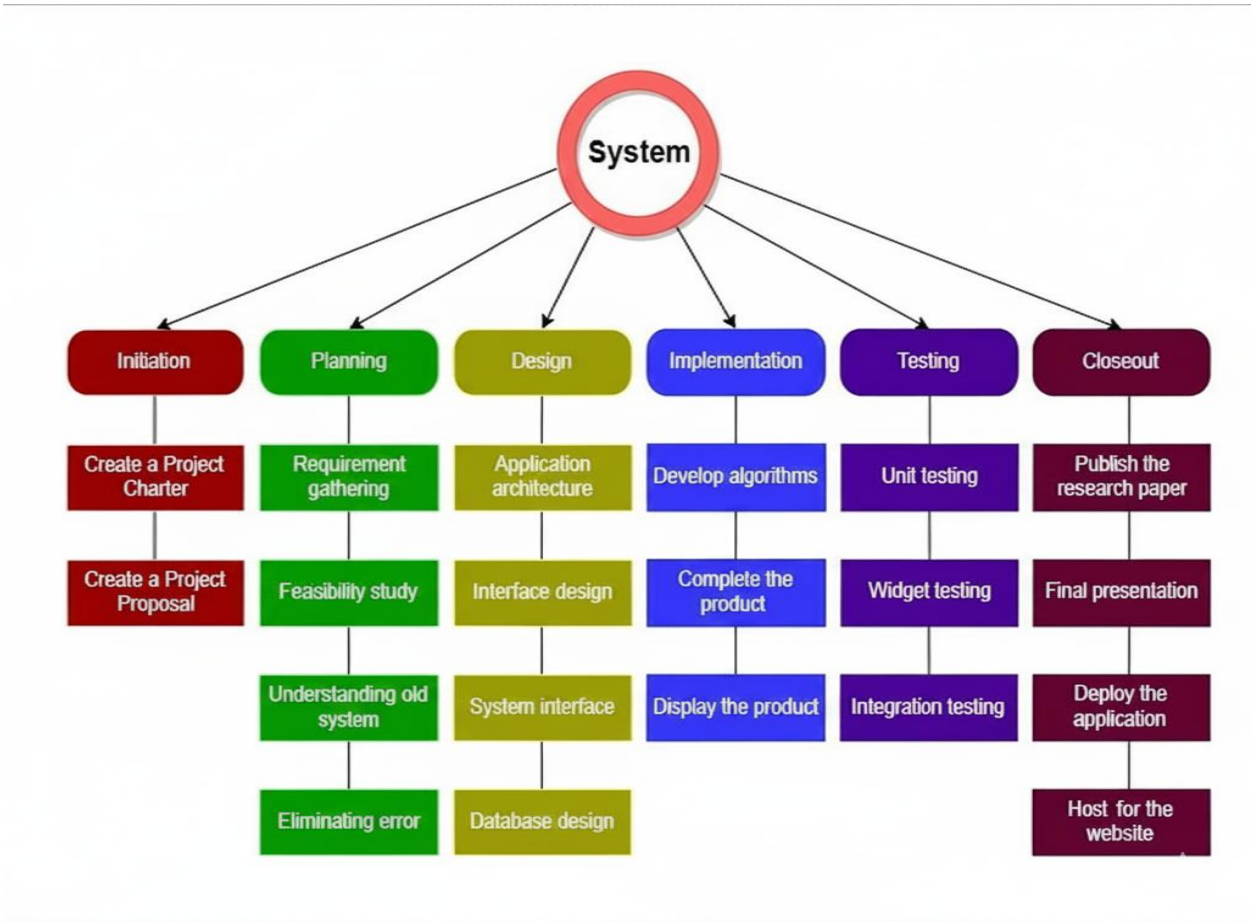
4.6 Hydration Management Component System Diagram



4.7 Grand Chart



4.8 Work Break down Structure



5.BUDGET AND BUDGET JUSTIFICATION

5.1 Budget

Cloud Services	LKR 40,000	Annual
MongoDB Atlas	LKR 15,000	Annual
Advertisement	LKR 1,500	Monthly
Development Tools	LKR 2,000	Monthly

5.2 Budget Justification

1.Cloud Services (LKR 40,000 annual):
The largest portion of the budget is allocated for cloud services like AWS, GCP, or Azure. These services provide essential resources such as compute power (EC2, Lambda), data storage, and other necessary cloud infrastructure to support the AI-driven hydration management system. The cloud services are crucial for scaling the system as user demand increases and for ensuring high availability and performance

2.MongoDB Atlas (LKR 15,000 annual):
MongoDB Atlas is selected for its efficiency in managing large volumes of unstructured data, such as user profiles, hydration history, and environmental data. The annual cost includes database hosting, backups, and security services, ensuring that the database is accessible, secure, and scalable.

3. Advertisement (LKR 1,500/month):
A portion of the budget is dedicated to social media advertising on platforms like Facebook and Instagram. This helps increase awareness of the platform after launch and allows for targeted audience to reach maximize user acquisition. These platforms offer affordable paid post boosting options that will drive traffic to the mobile application or website.

4.Development Tools (LKR 2,000 annual):
Minimal expenses are allocated for development tools and software libraries essential for coding and machine learning model development. This includes IDEs, version control services (like GitHub), and other utilities to support the backend and mobile app development.

5.3 Commercialization

1. Subscription Model

Overview: A subscription-based pricing model will be implemented to offer users various pricing tiers.
Basic Plan: Provides essential features, such as basic hydration tracking and general recommendations.

Premium Plan: Offers advanced features, such as personalized hydration plans, daily recommendations based on detailed activity logs, and in-depth health analytics.

2. Revenue Generation: Monthly or yearly subscriptions will generate recurring revenue, ensuring consistent cash flow and incentivizing users to stay engaged with the platform over time

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7.Appendices

Appendix A - Survey Link
https://docs.google.com/forms/d/e/1FAIpQLSeB-Pc4oz_n3VpxbqR08y9aBL1XOGqbeODFjj4Ns75bEyITsQ/viewform?usp=sharing&ouid=117357323975721066394